1. IDENTIFICATION

- Product Name: Pratico
- Main recommended use: insecticide of the neonicotinoid and triazol chemical group.

Registrant: **ADAMA BRASIL S/A**
Rua Pedro Antônio de Souza, 400 – Londrina – PR.
Parque Rui Barbosa. CEP 86031-610
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Emergency telephones: 0800 200 2345 – Adama Brasil / Toxiclin Serviços Médicos Ltda.
0800 722 6001 - RENACIAT (Rede Nacional de Centros de Informação e Assistência Toxicológica)
0800 400 7070 - SUATRANS COTEC

2. HAZARD IDENTIFICATION

- Most important hazards: the product may be toxic to humans and the environment IF NOT utilized per recommendations.
- Product Adverse Effects:

  Adverse effects to human health: the product is regarded as lethal if inhaled. Toxic IF ingested, and can be hazardous in contact with the skin. It can cause serious skin and eye irritation.

  Environmental effects: the product is considered hazardous to the aquatic organisms.

  Chemical and physical hazards: the product is not flammable.

- Main symptoms: the ingestion of this product may cause general symptoms like diarrhea, nausea, and vomit. The inhalation of the product dusts may cause the

irritation of the respiratory tract, respiratory insufficiency. The exposition to high doses of the product may stimulate SNC; provoke tremors, deliriums and disorientation. The direct contact with the skin, mucus and eyes can cause local irritation.

- Product hazard classification:

Hazard Classification System, according to the World Harmonized System for Classification and Labeling of Chemical Products. Norm ABNT-NBR 14725 – Part 2:2009.

Acute Toxicity - Oral: Category 4.
Acute Toxicity - Dermal: Category 5.
Acute Toxicity- Inhalation: Category 3.
Skin Corrosion/irritation: Category 2.
Serious eye damages / irritation: Category 2A.
Respiratory sensibilization: Impossible classification.
Skin sensibilization: Not classified.
Germ cell mutagenicity: Not classified.
Carcinogenicity: Impossible classification.
Toxicity to reproduction: Impossible classification.
Toxicity to organs-specific targets – Single exposition: Impossible classification.
Toxicity to organs-specific targets – Repeated exposure: Impossible classification.
Hazardous when inhaled: Impossible classification.
Hazardous to the aquatic environment – Acute: Category 3
Hazardous to the aquatic environment – Chronic: Impossible classification.
Flammable liquids: Not classified.

- Adequate labeling elements:

Pictogram		
Word of Warning	Danger	Attention

Hazard phrases:

H301 – Acute toxicity – Oral: Toxic if ingested.
H313 – Acute toxicity – Dermal: Can be hazardous in contact with the skin.
H330 – Acute toxicity – Inhaled: Fatal if inhaled.
H315 – Skin corrosion/Irritation: Causes skin irritation.
H319 – Serious eye harm/irritation: Cause serious eye irritation.
H402 – Hazardous to the aquatic environment - Acute: Hazardous to the aquatic organisms.

Precaution phrases:

P260 – Do not inhale the dusts/fumes/gases/mist/vapors/aerosol.
P264 – Wash thoroughly after handling.
P270 – Do not eat, drink or smoke during the handling of this product.
P271 – Use only in open air or well-ventilated places.
P280 – Wear protection gloves/clothes and eye/facial protection.
P284 – [In case of inadequate ventilation] Use respiratory protection equipment. P301 + P312 – IN CASE OF INGESTION: in case of indisposition, contact a CENTER FOR TOXICOLOGICAL INFORMATION/doctor.
302 + 352 – IN CASE OF SKIN CONTACT: Wash with plenty water and soap. P304 + P340 – IN CASE OF INHALATION: Remove the person to a well ventilated place and keep him/her at a restful position that allows for breathing.
P305 + P351 + P338 - IN CASE OF EYE CONTACT: Wash carefully with water for a long time. In case of eye-contact lenses, remove them IF possible. Continue rinsing.
P332 + P313 – In case of skin irritation or rash: Consult with a doctor.
P337 + P313 – In case the eye irritation persists: consult with a doctor.
P362 + P364 – Remove and wash all contaminated clothing before wearing it.
P330 – Rinse the mouth.
P403 + P233 – Store in a well ventilated place. Keep the container hermetically closed.
P405 – Store in a closed, locked storage.
P273 – Avoid the release to the environment.

3. INGREDIENTS' COMPOSITION AND INFORMATION

- Chemical nature: the product is a blend.
- Ingredients or impurities that may contribute to the hazard:

Chemical name	Nr CAS	Concen-tration	Molecular formula	Synonym	Hazard classification
(RS) – 2,4'difluoro-rethane-(1H-1,2,4triazol-1-ylmethyl) benzhydryl alchohol	7667421-0	200 g/L	$C_{16}H_{13}F_2N_3O$	-Flutriafol (flutriafol)	No data available
1-(6chloro-3pyridylmethyl)-Nnitroimidazolidin-2ylideneamine	13826141-3	250 g/L	$C_9H_{10}ClN_5O_2$	- Imidacloprido (imidacloprid)	- Acute toxicity – Oral: Category 4, -Hazard to aquatic environment – acute: Category 1.

Hazard classification system according to Globally Harmonized System for the Classification and Labeling of Chemical Products. Norm ABNT-NBR 14725 – Section 2:2009

4. FIRST AID MEASURES

- First Aid Measures: take the injured to an airy place. Remove the contaminated clothes. Wash the body parts contaminated with plenty of water and soap. If the injured is unconscious and can no longer breathe, apply the oxygenation or artificial respiration. Take the injured to the next medical service, together with this file.

- Inhalation: remove the person to a well-ventilated place. If the person is not breathing, apply the artificial respiration. If breathing with difficulty, consult with a doctor immediately. ATTENTION: Never feed an unconscious person via the mouth.

- Skin contact: immediately wash with plenty water and soap the affected area. Remove and wash the contaminated clothing before reusing, and discard the used contaminated. If symptoms/effects occur, consult with a doctor.

- Eye contact: wash them immediately with plenty of water for 15 minutes. Keep the eyelids open so that the proper eye rinsing is assured. If possible, remove the contact lenses. Consult with an eye doctor in case there´s an irritation.

- Ingestion: immediately wash the mouth with plenty of water. Do not induce vomit, but it is possible that it may occur spontaneously, and IF so, it should not be prevented. Lay the injured on the side so as to avoid breathing residues. Seek a doctor's attention immediately. ATTENTION: never feed an unconscious person via oral.

- Which actions must be avoided: do not perform mouth-to-mouth resuscitation IF the patient has ingested the product. Use an alternative or a device for the manual ventilation (type Ambu®) for the procedure.

- Protection for first-aid providers: avoid ingesting, inhaling, skin and eye contact with the product during the process.

- Notes to the doctor: there is no specific antidote. In case of ingestion of large amounts, apply the stomach pump and activated carbon. The treatment is symptomatic and shall include support measures such as hydroelectrolitic and metabolic disturbances. Monitor the hepatic and renal functions. In case of eye contact, apply the saline serum and lead to an eye examination.

5. MEASURES FOR FIRE-FIGHTING

- Adequate means of extinguishing: in case of fire, use water in form mist, CO_2 chemical powder.

- Means of extinguishing not recommended: avoid the use water jets directly on the product.

- Specific hazards and special methods for fire-fighting: non-flammable product. Evacuate the area and fight the fire at a safe distance. Use dykes to contain the water used in the extinguisher. Position yourself with the back to the wind. Use water in form of mist to cool the exposed equipment nearby the fire.

- Protection of people involved in the fire-fighting: Self-breathing apparatus adequate for fire-fighting.

- Specific hazards of the combustion of the chemical product: specific hazards of the product combustions are unknown.

6. CONTROL MEASURES FOR SPILL OR LEAK

- Personal precautions: utilize a waterproof overall, protection goggles, rubber boots and nitrile or PVC rubber gloves. The respiratory protection should be done depending on the present concentrations in the environment, or the extension of the spill/leak. Hence, one must choose semi facial or full facial masks with a replaceable filter cartridge, or even air purifiers with organic vapor filters.

  Removal of ignition sources: break the electric Power and turn off spark generators. Remove from the spot all materials that may cause any fires (ex.: diesel oil).

  Dust control: not applicable, since the product does not produce dust.

  Prevent inhalation and skin contact, mucosa and eyes: use clothing and accessories described above, under item Personal Care.

- Precautions for the environment: avoid contamination with waterways, blocking the entry of watercourse (sewer mouth). Prevent product residue to reach water sources.

- Methods for cleaning: in case of spill, stop the flow, not allowing the product to reach water drains and dumps, or bodies of water. Follow the instructions below: **Paved Flooring:** collect the material with the help of a shovel and place it in a container sealed and properly identified. The spilled product shall no longer be utilized. In this case, check with a registered company using the telephone in the label for its return and final destination. **Soil:** remove the layers of contaminated soil until you reach the clean soil; collect this material and place it in containers sealed and properly labeled. Contact the registered company as indicated above.
  Body of water: immediately interrupt the water collection for human and animal consumption, contact the environmental Office nearby and the company´s emergency center, since the measures to be taken depend on the size of the incident, the characteristics of the hydro body in questions, and the amount of product involved.

- Prevention of secondary hazards: avoid the product from contaminating rivers, streams, lakes, water sources, wells, sewers systems and effluents.

7. HANDLING AND STORAGE

- Handling:

- Technical measures: use the IPE (individual protection equipment) – Keep people, mainly children, and domestic animals away from the work spot. Do not come in direct contact with the product. Avoid spills or contamination of the equipment utilized during the supply.

  Prevention of worker´s exposition: use IPE according to description under item 8. Do not use damaged IPEs; do not use defected equipment, do not transport the product together with foods, medication, animal feed, and people.

  Precautions for safe handling: use IPEs according to description under item 8. Do not eat, drink or smoke during the handling and application of the product. Do not handle or apply the product without the recommended IPEs.

- Guidance for safe handling: use IPEs according to description under item 8. Handle the product with proper local exhaust, in well-ventilated area.

- Hygiene measures

Appropriate: take a shower immediately after the application of the product. Wash the contaminated clothing separately, avoiding contact with other personal use utensils. When washing the clothes, use gloves and water-proof apron.

Inappropriate: wash the contaminated clothes together with other clothing pieces or personal utensils.

- Storage

Appropriate technical measures: keep the product in its original packaging. Keep the eventual product leftovers in its original packaging, properly closed.

Inappropriate technical measures: avoid keeping the product next to heat sources and direct contact with sunlight.

Storage conditions

Adequate: keep the product in its original packaging, always closed. The storage spot must be exclusive for toxic products, must be isolated from foods, drinks, animal foods and other materials. The construction for storing must be of concrete or non combustible material. The place must be ventilated, covered and have impermeable flooring. Place a warning sign with the phrase: CAREFUL: POISON. Lock the place, blocking access to non authorized people, mainly children. There must always be available packaging to wrap the punctured packaging or to collect the leaked product. Observe the disposition in the Municipal and State Laws.

To avoid: humid and hot places.

Incompatible products and materials: do not store together with foods, drinks, including the ones destined to animals.

Materials safe for packaging

Recommended: product already in adequate packaging.

Inadequate: removing the product from its original packaging.

8. EXPOSITION CONTROL AND INDIVIDUAL PROTECTION

- Engineering control measures: use local exhaust and provide proper ventilation to the work place. The operator must always use respiratory equipment for protection, even when there is available good ventilation.

- Parameters for specific control:

Limits of occupational exposition:

Common name	Limit of Exposition	Type	Effect	References
Flutriafol	Not established	TLV-TWA	------	ACGIH 2011
		REL-TWA	------	NIOSH
		PEL-TWA	------	OSHA
Imidacloprid	Not established	TLV-TWA	------	ACGIH 2011
		REL-TWA	------	NIOSH
		PEL-TWA	------	OSHA

Biological indicators:

Common name	Collection time	Biological Limit	Type	Notes	References
Flutriafol	---	Not established	BEI	---	ACGIH 2011
Imidacloprid	---	Not established		---	ACGIH 2011

- Individual protection equipments:

Respiratory protection: use the combined mask with the chemical filter for organic vapors, acid gases and pesticides and mechanical particulate filter.

Hand protection: use the nitril rubber gloves, or other waterproof, tear and puncture resistant material.

Eye protection: use safety goggles for chemical products.

Body and skin protection: use the long sleeve waterproof, or hydro repellent overall and PVC boots.

- Special precautions: keep the IPEs properly clean and in adequate use conditions, by periodically inspecting them, and doing the possible maintenance and/or replacement of damaged equipment.

9. PHYSIOCHEMICAL PROPERTITES

- Physical state: liquid.
- Form: viscous.
- Color: Pantone Yellow C (basic color) and 7528 C (definitive color).
- Odor and odor limit: characteristic.
- pH: 8,83 + 0,20 to 1% (m/v) of Imidacloprid product 250 + Flutriafol 200 SC at temperature of 23ºC.
- Point of fusion: does not apply since it´s a liquid formulation product. Freezing point: not available.
- Initial boiling point and boiling range temperature: not available.
- Flash point: it was not possible to determine the flash point of the substance due to its water base and propylene glycol formulation. Since the water has a boiling point of approximately 100ºC and the propylene glycol has a flash point around 80ºC, it starts creating water vapor inside the glass, coming to extinguish the flame of the equipment.
- Rate of evaporation: not available.
- Flammabilty: not flammable.
- Inferior/superior Flammability or explosibility limit: Not explosive.
- Vapor pressure: not available.
- Vapor density: not available.

- Density: according to the result, the density of the test substance Imidacloprid 250 + Flutriafol 200 SC was 1,185 ± 0,002 g/cm^3 at temperature of 20ºC.
- Solubility/Miscibility: according to the results, the product Imidacloprid 250 + Flutriafol 200 SC, presented a homogenous mixture with water and separation from solid material with ethyl alcohol and hexane, at maximum dosage.
- Partition coefficient – n-octanol/water: not available.
- Self-ignition temperature: Auto/Self ignition: Based on the water content the product does not ignite.
- Decomposition temperature: not available.
- Viscosity: according to the results, the viscosity of the test substance, Imidacloprid 250 + Flutriafol 200 SC, was of 1300 + 60 cP at 20ºC.

- Corrosiveness: according to the results, the corrosiveness for the test substance Imidacloprid 250 + Flutriafol 200 SC, presented a corrosion rate of: Aluminum 1,0 x 10^{-4} ± 0,1 x 10^{-4} and iron 1,0 x 10^{-4} ± 0,1 x 10^{-4} mm/year. Copper and brass did not present a corrosiveness rate.
- Superficial tension: according to the results, the solution superficial tension at 1 % (m/v) of Imidacloprid 250 + Flutriafol 200 SC in water was 0,06660 Nm^{-1}.
- Oxidation: Not fire propagating

10. STABILITY AND REACTIVITY

- Reactivity: data not available.

- Chemical stability: according to the results the test substance was regarded as stable; due to the sample not vary more than 5% of the concentration of its active ingredient, when compared to a sample at ambient temperature.

- Possibility of hazardous reactions: there are no hazardous reactions known under normal conditions of use and storage.

- Conditions to be avoided: prolonged exposition to sunlight and direct exposition to the air with an open packaging.

- Incompatible materials: there is no available data.

- Hazardous decomposition products: there is no available data.

11. TOXICOLOGICAL INFORMATION

- Acute toxicity:

  DL_{50} Oral (female rats): > 300 mg/kg.
  DL_{50} Dermal (rats): > 4 000 mg/kg. (Determined value for Imidacloprid 250 + Flutriafol 200 SC).
  CL_{50} Inhalation (rats) (4h): > 0,612 mg/L (for the active ingredient Imidacloprid) and 0,593 mg/L (for the active ingredient Flutriafol).

- Local effects:

  Skin irritability: in test done with animals we reached an erythematic index 3/3 of the animals and edema in 2/3 of the animals. All the signs of irritation returned to a normal reading in 24 hours after the treatment. It´s considered as a light skin irritant. No behavioral or clinical alteration related to the treatment was observed during the period.

  Eye Irritability: in tests done on rabbits we got hyperemia and conjunctival edema in 3/3 of the eyes tested and conjunctival secretion in 2/3 of the eyes tested, and pericornean hyperemia in 1/3 of the eyes tested. All signs of irritation returned to normal in the reading 72 hours after the treatment. No behavioral or clinical alteration related to the treatment was observed during this time.

  Skin sensibilization: the epidermal application of Imidacloprid 250 + Flutriafol 200 SC did not cause dermal sensibilization in guinea pigs

  Respiratory sensibilization: there is no available data.

- Chronic toxicity:

  Mutagenicity in germinating cells: in tests done in *Salmonella typhimurium* coat and in Micronucleus test it showed the product is not mutagenic.

  Carcinogenicity: there is no available data.

  Toxicity to reproduction: there is no available data.

- Systemic toxicity to target organ:

Single exposition: there is no available data.

Repeat expositions: there is no available data.

Aspiration hazard: there is no available data.

- Main symptoms: the product ingestion may cause general symptoms such as nausea, vomit and diarrhea. The inhalation of the product dust may cause irritation in the respiratory tract, respiratory insufficiency. The exposition to the product in high dosages may stimulate the SNC, provoke tremors, delirium and disorientation. The direct contact with the skin, mucosa and eyes cause local irritation.

12. ECHOLOGICAL INFORMATION

- Environmental. Behavioral effects, and impacts of the product:

- Persistence and degradability: there is no available data.

- Ecotoxicity:

Toxicity to fish: CL_{50} (96h): 282,84 $mg.L^{-1}$. The greatest concentration that did not cause an acute lethal effect on fish was 200 $mg.L^{-1}$ and the smallest concentration that caused mortality to 100% of the fish population was 400 $mg.L^{-1}$, in 96 hours of exposition.

Toxicity for micro crustaceans: CE_{50} (48h): was not determined since the greater concentration tested was 100 $mg.L^{-1}$ of the product Imidacloprid 250 + Flutriafol 200 SC, did not cause immobility to the test organisms. Hence, the value CE50 of the substance test is greater than 100 $mg.L^{-1}$.

Toxicity to algae: CE_{50} (72h): 33,7 $mg.L^{-1}$.

Toxicity to soil organisms: CL_{50} (14 days): 21,02 $mg.kg^{-1}$ of artificial soil (confidence interval 95% of 18,34 to 24,10 $mg.kg^{-1}$). In 14 days, the greatest concentration tested that did not cause lethal effect was 6,25 $mg.kg^{-1}$ and the smallest concentration that caused mortality was at 100% was 50 $mg.kg^{-1}$ of artificial soil.

Toxicity for bees: DL_{50} (24 and 48h): were respectively 3,15 ug.bee^{-1} and 2,94 ug.bee^{-1}.

Toxicity to birds: The average lethal oral dosage (DL50) combined (males and females) after a single administration of Imidacloprid 250 + Flutriafol 200 SC on Japanese quail was estimated as 69,69 mg/kg of body weight, with confidence interval of 95%, according to the data analyzed of mortality-dosage using the Probit method.

Toxicity to microorganisms: to the final incubation period of 28 days, the difference between the control and soils treated was less than 25%, hence able to be regarded as significant. Based on the results of this study, the product may be regarded as not having a long term effect under the carbon transformation on the two types of soil analyzed in the present test. The deviations between control and soils treated for the transformation of nitrogen were under 25% (limit specified by the Guideline OECD 216 (2000)) and hence, regarded as not significant. Based on the results of this study, the product may be regarded as not having a significant long term effect under the transformation of nitrogen in the two soils evaluated in the present test.

- Bio-accumulative potential: there is no available data.

- Soil Mobility: there is no available data.

13. CONSIDERATION ON FINAL DESTINATION

- Treatment and disposition methods:

Product: disable the product via incineration in furnaces destined for this type of operation, equipped with effluent gases, approved by the competent office.

Product leftovers: keep the eventual remaining products; those with expired date in its original packaging adequately closed. The spot must be proper exclusively for toxic products, and must be isolated from foods, drinks or other materials. The place must be safe (covered, ventilated and with waterproof flooring).

Used packaging: the empty flexible packaging must be conditioned properly in a safe place (covered, ventilated and with waterproof flooring) for later return to the commercial establishment where it was purchased from, for one year, or in INPEV

collection centers. Do not burn, neither bury nor reutilize the packaging. The recycling may be applied as long as the pertinent laws are followed. Follow the specific Municipal and State Legislation. Consult the specific State and Municipal offices. Consult the State or Municipal Environmental Office.

14. INFORMATION ON TRANSPORTATION

• National and international Norms:

Classification:	UNO Nr:	UN2902
(Road Transportation)		
	Appropriate loading name:	**LIQUID PESTICIDE, TOXIC N.E. (flutriafol + imidacloprid)**
	Class of risk:	6.1
	Subclass of risk:	-
	Risk number:	60
	Packaging Group:	III
	Special provision:	61, 223, 274
Quantity limited by:	Vehicle:	333

	Internal packaging:	5 L
Packaging & IBCs:	Packaging Instructions:	P001 IBC03 LP01
	Special Provisions:	-
Tanks	Instructions:	T7
	Special Provisions:	TP2, TP28

IATA Classification: (Air Transport)	UN Number:	UN2902
	Proper name for loading:	**(LIQUID PESTICIDE, TOXIC N.E. (flutriafol + imidacloprid)**
	Class of risk:	6.1
	Subclass of risk:	-
	Risk number:	60
	Packaging Group:	III
	ERG Code	6L

Passenger and cargo aircraft:	As limited quantity, as packaging instructions:	Y642
	A limited quantity, maximum quantity per packaging:	2 L
	Packaging Instructions:	655
Cargo aircraft:	Maximum quantity per packaging:	60 L
	Packaging instructions:	663
	Maximum quantity per packaging:	220 L
	Special provision:	-

I.M.O. Classification:	UN number:	UN2902
(Maritime transportation)		

	Proper name for loading:	**(LIQUID PESTICIDE, TOXIC N.E. (flutriafol + imidacloprid)**
	Class of risk:	6.1
	Subclass of risk:	-
	Risk number:	60
	Packaging group:	III
	Marine pollutant:	Marine pollutant
	Special provision:	61, 223, 274
	Limited quantity:	5 L
	EmS:	F-A, S-A.
Packaging:	Instruction:	P001, LP01
	Provision:	-
IBC:	Instruction:	IBC03
	Provision:	-
Instruction for Tanks:	IMO:	-

	UN:	T7
	Provision:	TP2, TP28

15. INFORMATION ON NORMS

- Norms:
  ABNT NBR – 14725
  Norm 5232 – ANTT

16. OTHER INFORMATION

"This File was elaborated by TOXICLIN® Medical Services, from the data supplied by the registered Company. The information on this MSDS represent the present data and reflect exactly our Best knowledge for the appropriate handling of this product, according to the specifications in the label and leaflet. Any other use of the product other than recommended, shall be the responsibility of the user".

Acronyms:

ABNT – Brazilian Association of Technical Norms
ACGIH – *American Conference of Governmental Industrial Hygienists* **ANTT** – Terrestrial Transportation National Agency
BCF – Bioconcentration Factor **BEI** – Exposition Biological Index
CAS – *Chemical Abstracts Service*
CL_{50} – Lethal Concentration 50%
CE_{50} – Effective Concentration 50% **DL_{50}** – Lethal Dosage 50%
EPI – Individual Protection Equipment
GHS – Globally Harmonized System **for** Classification and Labeling of Chemical Products.
IARC – *Internacional Agency for Research on Cancer*
IATA – *International Air Transport Association*
ICAO – *International Civil Aviation Organization*

IMO – *Internacional Maritime Organization*
Koc – Organic Carbon-water in soil Partition Coefficient **Kow – ** N-octanol-water partition coefficient
Log Kow – N-octanol-water partition coefficient logarithm
NBR – Brazilian Norm
ND – Information not available for disclosure
NIOSH – *National Institute for Occupational Safety and Health* **NOEL** – Level where the effect is not observed
ONU – United Nations Organization
OSHA – *Occupational Safety & Health Administration*
PEL – *Permissible Exposure Limit*
REL – *Recommended Exposure Limit*
STEL – *Short Term Exposure Limit*
TLm – Tolerance Limit where 50% of the marine population tested presented abnormal behavior or death.
TLV – *Threshold Limit Value*
TWA – *Time Weighed Average*
WHO – *World Health Organization*

Legends:

Impossible Classification – there is no data sufficient or available for the classification of the product.
Not classified – product does not fit in the GHS classification category, hence, does not present risk.

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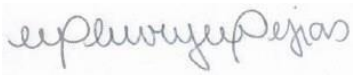
Done by: Michelle Fleury



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